

Basics of University Mathematics

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Linear Independence

Linear combinations of vectors can be used to represent points. For example, the point $(3, 2)$ can be expressed by the linear combination

$$3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

When the coefficients of the linear combination are arbitrary numbers, the linear combination can represent sets of points. For example, the linear combination

$$x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x, y \in \mathbb{R},$$

is known to represent all points in the space $\mathbb{R}^2$. Therefore, we have:

$$\left\{ \mathbf{z} \in \mathbb{R}^2 \mid \mathbf{z} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x, y \in \mathbb{R} \right\} = \mathbb{R}^2.$$

In fact, this can be done with almost any linear combination of two two-dimensional vectors.

Basic Exercise

- (a) Solve for which values of the coefficients x and y the linear combination

$$x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x, y \in \mathbb{R},$$

corresponds to the point $(7, 5)$. Verify your answer by substituting the solved values of the unknowns x and y and simplifying the resulting expression.

- (b) Solve for which coefficients the linear combination

$$x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x, y \in \mathbb{R},$$

corresponds to the point (a, b) . Verify your answer by substituting the solved values of the unknowns x and y .

Part (b) of the previous problem implies that any point in the space $\mathbb{R}^2$ can be expressed as a linear combination of these vectors. Therefore, we have:

$$\left\{ \mathbf{z} \in \mathbb{R}^2 \mid \mathbf{z} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix}, x, y \in \mathbb{R} \right\} = \mathbb{R}^2.$$

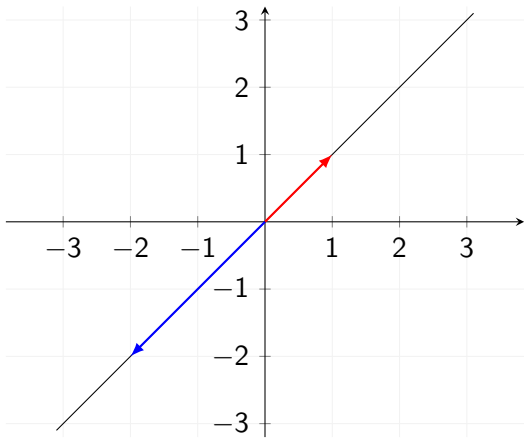
However, representing all points in the plane $\mathbb{R}^2$ cannot be achieved with the linear combination of a single vector $x\mathbf{v}$, $x \in \mathbb{R}$, as this linear combination corresponds to a line (passing through the origin) and can only represent the points on that line. Similarly, linear combinations of two or more vectors can represent a line if some of the vectors are redundant in the linear combination. For example, in the linear combination

$$x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} -2 \\ -2 \end{bmatrix}, \quad x, y \in \mathbb{R},$$

one of the vectors is redundant because it can be simplified away:

$$x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} -2 \\ -2 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-2)y \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (x - 2y) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The expression $x - 2y$ could be replaced with an arbitrary coefficient t , because if x and y are arbitrary real numbers, then $x - 2y$ can also take arbitrary values. The accompanying image illustrates the vectors of the previous linear combination as well as the set of points defined by the linear combination.



Basic Exercise

Simplify the following linear combinations to a linear combination of a single vector:

$$(a) \ x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} -3 \\ -6 \end{bmatrix} + z \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}, \ x, y \in \mathbb{R},$$

$$(b) \ x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \ x, y \in \mathbb{R},$$

Thus, the number of vectors is not the only important factor in a linear combination, but also the nature of the vectors. If none of the vectors are redundant in the linear combination, the vectors are said to be **linearly independent**. If, however, one of the vectors is redundant, the vectors are said to be **linearly dependent**. Two-dimensional vectors can only be linearly dependent if one vector is a multiple of another or if one is the zero vector. In the three-dimensional case, this can happen in other ways, which will be explored next.

Let's first show that with the linear combination

$$x \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad x, y, z \in \mathbb{R},$$

we can represent all points in three-dimensional space $\mathbb{R}^3$. Let (a, b, c) be an arbitrary point in this space. Then, we have:

$$x \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \iff \begin{bmatrix} x+z \\ x+y \\ y+z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \iff \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

and further,

$$\begin{bmatrix} 1 & 0 & 1 & | & a \\ 1 & 1 & 0 & | & b \\ 0 & 1 & 1 & | & c \end{bmatrix} \iff \dots \iff \begin{bmatrix} 1 & 0 & 0 & | & \frac{1}{2}(a+b-c) \\ 0 & 1 & 0 & | & \frac{1}{2}(-a+b+c) \\ 0 & 0 & 1 & | & \frac{1}{2}(a-b+c) \end{bmatrix}$$

$$\iff \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} a+b-c \\ -a+b+c \\ a-b+c \end{bmatrix}.$$

Thus, any point can be represented by this linear combination. (How would you express, for example, the point $(1, 2, 3)$ in standard coordinates using this linear combination?) Let's verify:

$$\frac{1}{2}(a+b-c) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{1}{2}(-a+b+c) \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{2}(a-b+c) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \cdots = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

On the other hand, with the vectors

$$x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}, \quad x, y, z \in \mathbb{R},$$

we cannot represent every point in the space $\mathbb{R}^3$, even though none of the vectors is a multiple of another.

This attempt leads to the situation

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & a \\ 0 & 1 & 1 & b-a \\ 0 & 0 & 0 & \frac{1}{2}(c-b) \end{array} \right],$$

so, for example, representing the point $(0, 0, 1)$ with this linear combination is not possible (How is this reflected in the augmented matrix?).

This happens because there is a redundant vector in the set of vectors in terms of the linear combination. Thus, the vectors are linearly dependent. But why does this occur now, even though the vectors are not multiples of one another? The reason becomes clear from the geometric illustration (Draw using GeoGebra 3D, for example): the vectors lie on the same plane (passing through the origin), so their linear combination cannot extend beyond that plane.

Since this plane can be described by just two vectors lying within the plane, one vector in the linear combination must be redundant. This can also be seen algebraically:

$$\begin{aligned}x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} &= x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \left(2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \\&= (x + 2z) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + (y + z) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\&= s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad s, t \in \mathbb{R}.\end{aligned}$$

Thus, the last vector could be written as a linear combination of the other two, making it redundant for the linear combination. This also means, of course, that any one of the three vectors can be written as a linear combination of the others.

Basic Exercise

Write the vector $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ as a linear combination of the vectors $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$ and $\begin{bmatrix} 3 & 4 & 5 \end{bmatrix}^T$, and the vector $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$ as a linear combination of the vectors $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ and $\begin{bmatrix} 3 & 4 & 5 \end{bmatrix}^T$.

In the case of three-dimensional vectors, a redundant vector may exist in the linear combination even when no vector is a multiple of another, and this happens when one vector can be expressed as a linear combination of the others. This concept also generalizes to vectors in the space $\mathbb{R}^n$.

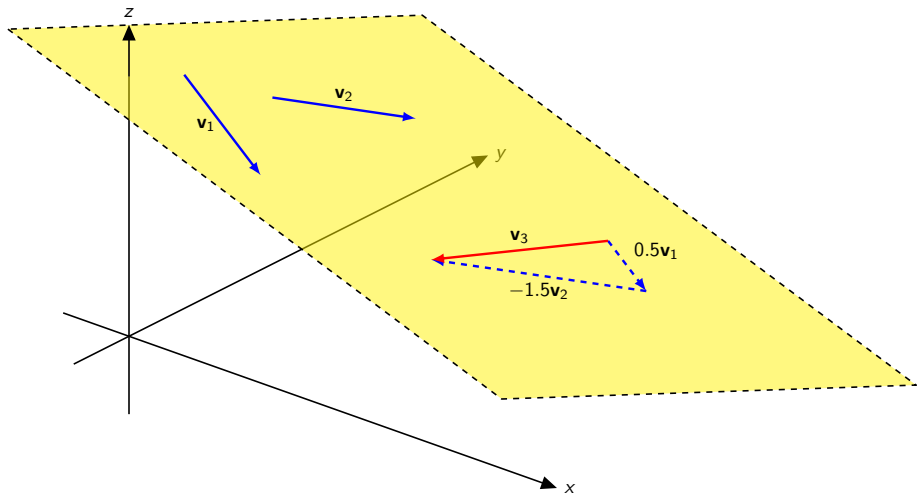


Figure: Linearly dependent three dimensional vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

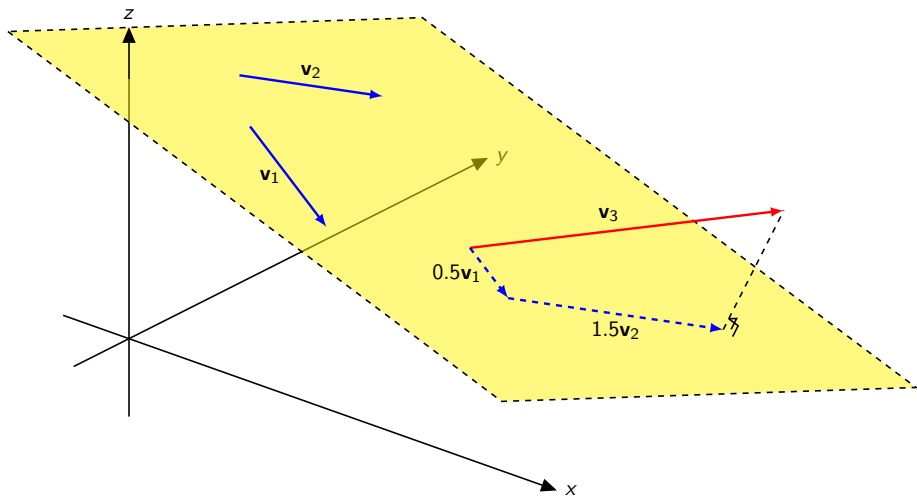


Figure: Linearly independent three dimensional vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

Definition of Linear Independence

Earlier, it was introduced that the linear independence of a set of vectors **specifically** means that there are no vectors in the set that are redundant in terms of linear combinations. This was explored by testing whether all points in the given space can be represented as a linear combination of the vectors. However, this technique only works for space $\mathbb{R}^n$ when there are exactly n vectors. For example, the vectors $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$ and $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$ are linearly independent, even though they cannot represent all points in the space $\mathbb{R}^3$. On the other hand, even when this technique is applicable, there is often a more convenient approach, as will be seen shortly.

Previously, linear dependence among vectors was deduced by expressing one of the vectors as a linear combination of others. This can also be done more easily. Next, we present the definition of linear independence, that is, what it **exactly means**. By using this definition, one can straightforwardly test the linear independence/dependence of a set of vectors in any case. Since the content of the definition is not entirely self-evident, we will derive it using the concept of linear dependence.

A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is linearly dependent if at least one of its vectors can be expressed as a linear combination of the others, which means that

$$\mathbf{a}_i = b_1\mathbf{a}_1 + b_2\mathbf{a}_2 + \dots + b_{i-1}\mathbf{a}_{i-1} + b_{i+1}\mathbf{a}_{i+1} + \dots + b_n\mathbf{a}_n,$$

where at least one coefficient b_i is non-zero. The same idea can be stated with all terms on one side of the equation: a set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is linearly dependent if

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_{i-1}\mathbf{a}_{i-1} + c_i\mathbf{a}_i + c_{i+1}\mathbf{a}_{i+1} + \dots + c_n\mathbf{a}_n = \mathbf{0}$$

holds, with at least one coefficient c_i non-zero. Thus, the only case in which the vectors are not linearly dependent is when the previous equation holds only when all coefficients are zero. This is the definition of linear independence.

A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is **linearly independent** if the following holds

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_n\mathbf{a}_n = \mathbf{0} \implies c_1 = c_2 = \dots = c_n = 0. \quad (1)$$

If the set of vectors is not linearly independent, it is **linearly dependent**.

The implication of the definition can be written more compactly as $A\mathbf{c} = \mathbf{0} \implies \mathbf{c} = \mathbf{0}$, where $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ and $\mathbf{c} = [c_1 \ c_2 \ \cdots \ c_n]^T$. This implication means that the only possible solution to the system of equations on the left-hand side of the implication is $\mathbf{c} = \mathbf{0}$.

(This does not yet confirm that this solution satisfies the equation; this is shown by the reverse direction $A\mathbf{c} = \mathbf{0} \iff \mathbf{c} = \mathbf{0}$, which is trivially true. Together, these implications mean that the system of equations must have a unique solution $\mathbf{c} = \mathbf{0}$. The direction $\iff$ does not need to be considered when determining linear independence, as it is obviously always satisfied.)

Example

We will show based on the definition that the vectors $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T$, $\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T$, and $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$ are linearly independent. It must be shown that the system of equations

$$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has the unique solution $c_1 = c_2 = c_3 = 0$.

Let us start by writing the system of equations as an augmented matrix

$$\begin{aligned} c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} &\iff \begin{bmatrix} c_1 + c_3 \\ c_1 + c_2 \\ c_2 + c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ &\iff \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \iff \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]. \end{aligned}$$

Example (continued)

Next, we solve the system of equations in this form:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \iff \dots \iff \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \iff \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

so the system of equations has the unique solution $\mathbf{c} = \mathbf{0}$. Therefore, the given vectors are linearly independent.

Example

We will show based on the definition of linear independence that the vectors $[1 \ 1 \ 1]^T$, $[1 \ 2 \ 3]^T$, and $[3 \ 4 \ 5]^T$ are linearly dependent. Solving the system of equations $\mathbf{A}\mathbf{c} = \mathbf{0}$ leads to the form

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right],$$

so the solution is not unique. Thus, the vectors are linearly dependent.

Example (continued)

From the solution of the system of equations

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = c_3 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \quad c_3 \in \mathbb{R}$$

we can deduce the values of the coefficients that express the vectors in the set as a linear combination of each other. For instance, in the case $c_3 = -1$, we obtain the example used in the derivation

$$\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Basic Exercise (★)

Are the vectors $\mathbf{a} = [1 \ 1 \ 1]^T$, $\mathbf{b} = [1 \ 0 \ 1]^T$, and $\mathbf{c} = [1 \ 0 \ 0]^T$ linearly dependent or independent?

If the matrix formed from the vectors in question happens to be a square matrix, the linear independence/dependence can conveniently be checked using the determinant.

The columns of a square matrix A are linearly independent if and only if $\det(A) \neq 0$.

The previous result can be straightforwardly justified using the inverse matrix (Justify).

Example

Are the following sets of vectors linearly independent or dependent:

$$(a) \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$(b) \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

$$(c) \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\}?$$

Solution

- (a) Since the coefficient matrix of the system of equations is a square matrix, the matter can be resolved using the determinant:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot (1 \cdot 1 - 0 \cdot 0) - 0 + 0 = 1 \neq 0,$$

thus the vectors are linearly independent.

- (b) Since the coefficient matrix of the system of equations is not a square matrix, the determinant cannot be used; we must resort to the definition.

$$\begin{aligned} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \iff \begin{bmatrix} 1 & 1 & | & 0 \\ 1 & 2 & | & 0 \\ 1 & 3 & | & 0 \end{bmatrix} \\ \iff \dots \iff \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} &\iff \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \end{aligned}$$

thus the vectors are linearly independent.

Solution (continued)

- (c) The vectors are clearly multiples of each other, meaning they can be expressed as linear combinations of each other. Therefore, the vectors are linearly dependent.

Subspace, Span, Basis, and Basis Vectors

In the introduction, it was noted that linear combinations of vectors can represent sets of points. It is said that the vectors of a linear combination **span** that set. This set can be the entire space or a subset of it. For example, in three-dimensional space, 2 linearly independent vectors span a plane through the origin, 1 linearly independent vector spans a line through the origin, and the zero vector spans the origin. These sets are called the **subspaces** of the space $\mathbb{R}^3$.

The set

$$\{ \mathbf{y} \mid \mathbf{y} = c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \cdots + c_n \mathbf{a}_n, c_i \in \mathbb{R} \}$$

is called the **subspace** spanned by the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ in $\mathbb{R}^n$.

In fact, based on the precise definition of a subspace, which is not covered in this course, $\mathbb{R}^n$ itself is also a subspace. Note that subspaces always pass through the origin. How can this be concluded from the previous linear combination?

It is important to note that the spanning vectors of a subspace can also include vectors that are redundant in terms of the linear combination.

If the set of vectors spanning a subspace is linearly independent, it is called the **basis** of the subspace, and its vectors are called the **basis vectors** of the subspace.

Thus, the basis contains the minimum possible number of vectors needed to span the subspace.

The **dimension** of a subspace refers to the number of basis vectors of the subspace.

Example

The vectors $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$, $\begin{bmatrix} 2 & 2 & 2 \end{bmatrix}^T$, and $\begin{bmatrix} 3 & 3 & 3 \end{bmatrix}^T$ in $\mathbb{R}^3$ span the subspace

$$\left\{ \mathbf{y} \mid \mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}, c_1, c_2, c_3 \in \mathbb{R} \right\} = \left\{ \mathbf{y} \mid \mathbf{y} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, t \in \mathbb{R} \right\},$$

which has a dimension of 1. This is a line passing through the origin.

The vectors $\begin{bmatrix} 2 & 3 \end{bmatrix}^T$, $\begin{bmatrix} 4 & 6 \end{bmatrix}^T$, and $\begin{bmatrix} -6 & -9 \end{bmatrix}^T$ in $\mathbb{R}^2$ span the subspace

$$\left\{ \mathbf{y} \mid \mathbf{y} = c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} -6 \\ -9 \end{bmatrix}, c_1, c_2, c_3 \in \mathbb{R} \right\} = \left\{ \mathbf{y} \mid \mathbf{y} = t \begin{bmatrix} 2 \\ 3 \end{bmatrix}, t \in \mathbb{R} \right\},$$

which also has a dimension of 1. Again, this is a line passing through the origin.

The next point was probably evident from the introduction already. However, let's write it down clearly as well.

You need n linearly independent vectors of $\mathbb{R}^n$ to span the space $\mathbb{R}^n$.

Basic Exercise

Let's continue examining the vectors from Exercise (★).

- (a) Can every point in space $\mathbb{R}^3$ be represented as a linear combination of them?
- (b) Do they span the space $\mathbb{R}^3$?
- (c) Repeat the previous questions, considering only the vectors **a** and **b**.

Exercise

- (a) What kind of geometric set do two non-zero vectors in $\mathbb{R}^2$ span if they are (i) linearly dependent (ii) linearly independent?
- (b) How many vectors in $\mathbb{R}^2$ can maximally be in a linearly independent set? What about in $\mathbb{R}^n$?
- (c) If we randomly select two vectors from $\mathbb{R}^2$, are they more likely to be linearly dependent or independent?
- (d) Justify without solving the system of equations and without using the determinant that the vectors $\begin{bmatrix} 2 & 2 \end{bmatrix}^T$ and $\begin{bmatrix} -1 & -1 \end{bmatrix}^T$ are linearly dependent.
- (e) What kind of geometric set do three non-zero vectors in $\mathbb{R}^3$ span if they are:
 - (i) linearly independent
 - (ii) linearly dependent, but each pair of vectors is linearly independent
 - (iii) linearly dependent and each pair of vectors is also linearly dependent?
- (f) Justify without solving the system of equations and without using the determinant that the vectors $\begin{bmatrix} 2 & 1 & 2 \end{bmatrix}^T$, $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$, and $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$ are linearly dependent.

Exercise

Let's examine the vectors $\mathbf{a} = [1 \ 1 \ 1]^T$, $\mathbf{b} = [1 \ 0 \ 1]^T$, and $\mathbf{c} = [1 \ 0 \ 0]^T$.

- (a) Express the point $(1, 2, 3)$ in $\mathbb{R}^3$ as a linear combination of the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
- (b) What are the coordinates of that point in a coordinate system aligned with the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ (instead of the standard basis vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$)?
- (c) Express the point (x, y, z) in $\mathbb{R}^3$ as a linear combination of the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
- (d) Based on the previous part, do the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ span the space $\mathbb{R}^3$?

Advanced Concepts

Standard basis vectors are orthogonal to each other and have a length of one. It might seem natural that this should apply to all basis vectors, but for basis vectors, it is sufficient that they are linearly independent. For this reason, orthogonal basis vectors of unit length have special names.

A basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ of the space $\mathbb{R}^n$ is called **orthogonal** if all its vectors are orthogonal to each other. The basis is called **orthonormal** if, in addition to orthogonality, the lengths of the vectors are equal to one.

Orthogonal means perpendicular or right-angled.

There is also a shorthand notation for the subspace spanned by vectors.

We use notation

$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{\mathbf{y} \mid \mathbf{y} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \dots + c_n\mathbf{a}_n, c_i \in \mathbb{R}\}$
for the subspace spanned by the set $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$.

As mentioned earlier, a function can be associated with a matrix $A \in \mathbb{R}^{m \times n}$, specifically the mapping $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, and sometimes the matrix is also used as the name of this function in place of the letter $\mathbf{f}$. In this case, one can speak of the range of matrix A , which is actually the set formed by all possible linear combinations of the columns of the matrix.

Let $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$. Then the range of the matrix is

$$\begin{aligned}\mathcal{R}(A) &= \{ \mathbf{y} \mid \mathbf{y} = A\mathbf{x}, \mathbf{x} \in \mathbb{R}^n \} \\ &= \{ \mathbf{y} \mid \mathbf{y} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n, x_i \in \mathbb{R} \}.\end{aligned}$$

This is called the **column space** of the matrix A , and it is the subspace spanned by the columns of the matrix.

Let $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$. Then

$$\mathcal{R}(A) = \text{span} \{ \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \}.$$

Rank of a Matrix

Basic Exercise

Using the definition of linear independence and Gaussian elimination, determine whether the columns of the following matrices are linearly dependent or independent, and express the linear combinations of the columns in the simplest form possible:

$$(a) A = \begin{bmatrix} 1 & -2 & 3 \\ -3 & 6 & -9 \\ 2 & -4 & 6 \end{bmatrix}$$

$$(b) B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 3 & -2 & 2 \end{bmatrix}$$

$$(c) C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 3 & 4 & 9 \end{bmatrix}.$$

The matrices A and C from the previous task are similar in the sense that the columns of both are linearly dependent. However, they are still essentially different; namely, from the simplified form of the linear combination, it can be observed that the columns of matrix A are “more linearly dependent” since two of them are redundant in terms of linear combinations. This property is measured by a concept called the rank of a matrix.

The **rank** of a matrix A refers to the number of its linearly independent columns. It is denoted by $\text{rank}(A)$.

By comparing the concepts of the rank and the range of a matrix, we notice the following connection.

Let A be a matrix. Then

$$\dim(\mathcal{R}(A)) = \text{rank}(A).$$

This could also be used as the definition of rank.

Basic Exercise

Continue analyzing the matrices from the previous task.

- (a) Determine the ranks of the matrices based on simplified linear combinations.
- (b) Next, consider the transposes of the matrices A^T , B^T , and C^T .
 - (i) Determine the linear independence/dependence of the columns of the matrices based on the definition and using Gaussian elimination.
 - (ii) Express the linear combinations of the columns of the matrices in the simplest form possible.
 - (iii) Determine the ranks of the matrices.

From the previous task, it is observed that the matrices examined behave in terms of linear independence and rank in exactly the same way, regardless of whether one considers their columns or rows. This is not a coincidence, as it generally holds true that

$$\text{rank}(A) = \text{rank}(A^T).$$

Thus, a matrix has as many linearly independent rows as it has linearly independent columns. Whether one is interested in the number of linearly independent row or column vectors of the matrix depends on context.

In the above, the rank of the matrix has been defined by expressing the linear combination of its columns in the simplest form possible and counting how many vectors remain. This can be more conveniently done based on the row echelon form of the matrix.

The rank of a matrix is obtained from its row echelon form as the number of non-zero rows.

Verify that this is true for the previous matrices. (This is the reason why the tasks were instructed to examine the matter precisely through the definition and Gaussian elimination).

Since the rank of a matrix is related to the number of zero rows in its row echelon form, and the number of zero rows is related to the solutions of the corresponding system of equations, the rank of a matrix has a connection to the solutions of the corresponding system of equations. This will be explored in the next chapter.

General Properties of Solutions

A linear system of equations

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \right. , \quad (2)$$

that is, the equation $A\mathbf{x} = \mathbf{b}$

- has solutions if and only if $r = \text{rank}(A) = \text{rank}(\tilde{A})$
- has exactly one solution if $r = n$
- has infinitely many solutions if $r < n$.

Basic Exercise

Form a 2×2 system of equations such that

- (a) there is exactly one solution
- (b) there are no solutions
- (c) there are infinitely many solutions.

and determine $\text{rank}(A)$, $\text{rank}(\tilde{A})$, and n in each case. Verify your answers by solving the systems you created.

For a homogeneous linear system of equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases} \quad (3)$$

the following holds:

- The **trivial solution** $\mathbf{x} = \mathbf{0}$ always exists.
- The trivial solution is the only solution if $r = n$.
- Non-trivial solutions exist if $r = \text{rank}(A) < n$, in which case the solution is multi-valued.
- In the case $r < n$, the solutions form a space passing through the origin, called the **null space** or **kernel**, and its dimension, or the number of its basis vectors, is $n - r$. The notations for null space are $\mathcal{N}(A)$, $\text{null}(A)$ and $\ker(A)$.
- The dimension of the null space is called the **nullity** of the coefficient matrix A and is denoted by $\text{nullity}(A)$.
- For the range and null space of a matrix we have

$$\dim(\mathcal{R}(A)) + \dim(\text{null}(A)) = n$$

or equivalently

$$\text{rank}(A) + \text{nullity}(A) = n.$$

This is known as the **rank-nullity theorem**.

In practice, the nullity of A represents the number of free variables in the solution of the system $A\mathbf{x} = \mathbf{b}$ (in the case of a non-homogeneous system, under the assumption that solutions exist).

Example

Consider the system $A\mathbf{x} = \mathbf{0}$. Let B be the matrix obtained by transforming matrix A into row echelon form using Gaussian elimination. Determine $\text{rank}(A)$, $\text{nullity}(A)$, and n .

$$(a) B = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$(b) B = \begin{bmatrix} 2 & 4 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$(c) B = \begin{bmatrix} 2 & 4 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(d) B = \begin{bmatrix} 2 & 4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(e) B = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solution

The rank of matrix A is obtained as the number of linearly independent rows (or columns) of matrix B .

n is the number of variables in the system, which can be inferred from the size of matrix B , based on the fact that the product $A\mathbf{x}$, and thus also the product $B\mathbf{x}$, must be defined.

After this, the nullity of matrix A is obtained from the formula $\text{nullity}(A) = n - \text{rank}(A)$. The dimension of the null space can of course also be inferred by solving the system of equations and finding the number of free parameters.

(a) $\text{rank}(A) = 3$, and since $B \in \mathbb{R}^{3 \times 3}$, it must hold that $\mathbf{x} \in \mathbb{R}^{3 \times 1}$, hence $n = 3$. Thus, we have $\text{nullity}(A) = 3 - \text{rank}(A) = 0$, meaning

(b)

$$\text{rank}(A) = 3,$$

$$A \in \mathbb{R}^{3 \times 4} \implies \mathbf{x} \in \mathbb{R}^{4 \times 1} \implies n = 4,$$

$$\text{nullity}(A) = 4 - 3 = 1$$

Solution (continued)

(c)

$$\text{rank}(A) = 3,$$

$$A \in \mathbb{R}^{4 \times 5} \implies \mathbf{x} \in \mathbb{R}^{5 \times 1} \implies n = 5,$$

$$\text{nullity}(A) = 5 - 3 = 2$$

(d)

$$\text{rank}(A) = 1,$$

$$A \in \mathbb{R}^{3 \times 2} \implies \mathbf{x} \in \mathbb{R}^{2 \times 1} \implies n = 2,$$

$$\text{nullity}(A) = 2 - 1 = 1,$$

so the solutions form a line in $\mathbb{R}^2$.

(e)

$$\text{rank}(A) = 2,$$

$$A \in \mathbb{R}^{4 \times 3} \implies \mathbf{x} \in \mathbb{R}^{3 \times 1} \implies n = 3,$$

$$\text{nullity}(A) = 3 - 2 = 1,$$

so the solutions form a line in $\mathbb{R}^3$.

Next, let's take a corresponding example for the case of a non-homogeneous system of equations.

Example

Let

$$B = \left[\begin{array}{cccc|c} 2 & 4 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 \end{array} \right]$$

be the matrix obtained when the augmented matrix $\tilde{A} = [A \ \mathbf{b}]$ associated with the system $A\mathbf{x} = \mathbf{b}$ is transformed into row echelon form using Gaussian elimination. We want to determine how many free parameters remain in the solution of the system $A\mathbf{x} = \mathbf{b}$ without solving the system.

It was previously stated that $\text{nullity}(A)$ also indicates the number of free parameters in the solution of a non-homogeneous system, provided that solutions exist. Thus, it suffices to show that solutions exist and to determine $\text{nullity}(A)$.

Since $\text{rank}(A) = 3 = \text{rank}(\tilde{A})$, solutions exist. Additionally, $n = 4$, so we have $\text{null}(A) = 4 - 3 = 1$. Thus, there remains one free parameter in the solution.

Basic Exercise

Let B be the matrix obtained when the augmented matrix $\tilde{A} = [A \ \mathbf{b}]$ associated with the system $A\mathbf{x} = \mathbf{b}$ is transformed into row echelon form using Gaussian elimination. Determine $\text{rank}(A)$, $\text{nullity}(A)$, and the number of free variables in the solution of the system $A\mathbf{x} = \mathbf{b}$, when

$$(a) \ B = \left[\begin{array}{ccccc|c} 2 & 4 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -3 \\ 0 & 0 & 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(b) \ B = \left[\begin{array}{cc|c} 2 & 4 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$(c) \ B = \left[\begin{array}{ccc|c} 2 & 4 & 0 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Basic Exercise

Form a homogeneous system of equations with three variables,

- (a) that has exactly one solution
- (b) that has no solutions
- (c) whose solution has 1 free parameter (What size can the coefficient matrix be?)
- (d) whose solution has 2 free parameters (What size can the coefficient matrix be?),

and determine $\text{rank}(A)$, $\text{nullity}(A)$, and n in each case. Verify your answers by solving the systems you created.

Cramer's Rule

Cramer's rule can be used in certain situations to solve a system of linear equations by utilizing determinants: If the $n \times n$ system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n = b_n \end{cases} \quad (4)$$

has a non-zero determinant of the coefficient matrix, that is, $D = \det(A) \neq 0$, the system has exactly one solution, which is given by the formulas

$$x_1 = \frac{D_1}{D}, \quad x_2 = \frac{D_2}{D}, \quad \cdots, \quad x_n = \frac{D_n}{D}, \quad (5)$$

where D_k is the determinant obtained from D by replacing the k^{th} column with the elements $b_1, \cdots, b_n$.

As a consequence of Cramer's rule, we obtain a formula for calculating the inverse matrix using determinants:

$$A^{-1} = \frac{1}{\det(A)} [C_{jk}]^T = \frac{1}{\det(A)} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \quad (6)$$

where C_{jk} is the cofactor corresponding to a_{jk} . In the case of a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the formula for the inverse matrix takes the form

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \quad (7)$$

This is quite simple; however, unfortunately, for larger matrices, the formula is too cumbersome for manual calculation. For example, in the case of a 3×3 matrix

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

the formula is given by

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} & -\begin{vmatrix} b & c \\ h & i \end{vmatrix} & \begin{vmatrix} b & c \\ e & f \end{vmatrix} \\ -\begin{vmatrix} d & f \\ g & i \end{vmatrix} & \begin{vmatrix} a & c \\ g & i \end{vmatrix} & -\begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ \begin{vmatrix} a & e \\ g & h \end{vmatrix} & -\begin{vmatrix} a & b \\ g & h \end{vmatrix} & \begin{vmatrix} a & b \\ c & e \end{vmatrix} \end{bmatrix}.$$

Except for the 2×2 case, it is advisable to compute the inverse matrix using the Gaussian algorithm instead of the formula (6). However, it is worth remembering that formula (6) is of the form “one over the determinant times some matrix,” as this clearly indicates that the inverse matrix does not exist if the determinant of the matrix is zero.

Cramer's rule is sometimes convenient for manual calculations and can be easily applied even if matrix A contains parameters (i.e., some or even all elements are unknown). It also does not involve divisions, which can often help with numerical calculations. Additionally, when deriving theoretical results, Cramer's rule can significantly simplify intermediate steps. However, when calculating how many operations are needed for the expansion of determinants, it is observed that this number is so large that this method is not often used in practical applications.

Basic Exercise

Calculate the inverse matrix of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

using both the standard formula and the Gaussian algorithm.

Geometric Interpretation of the Determinant

In the context of the cross product, it was noted that the volume V of the parallelepiped spanned by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ can be expressed using the determinant in the form

$$V = \left| \det \begin{pmatrix} \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \end{pmatrix} \right|,$$

where the vertical bars denote the absolute value. Based on property (1) of determinants, this can also be written as

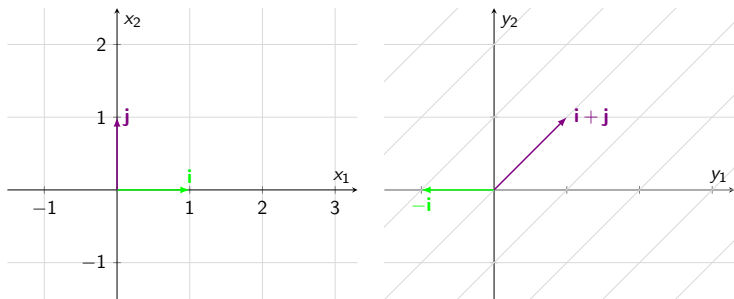
$$V = \left| \det \begin{pmatrix} \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} \end{pmatrix} \right| = |\det([\mathbf{u} \ \mathbf{v} \ \mathbf{w}])|.$$

This interpretation generalizes to n dimensions, meaning that $|\det(A)|$ generally represents the volume of the n -dimensional parallelepiped spanned by the columns (or rows) of a square matrix. This leads to an interesting interpretation of the determinant in the context of coordinate transformations.

Let's first consider the case of two variables. In Chapter 4.3.1, it was noted that multiplying by the matrix

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$$

corresponds to a transformation to the coordinate system spanned by the vectors $\begin{bmatrix} -1 & 0 \end{bmatrix}^T$ and $\begin{bmatrix} 1 & 1 \end{bmatrix}^T$. In this case, the parallelogram (now a square) spanned by the original basis vectors $\mathbf{i}$ and $\mathbf{j}$ is transformed into a parallelogram spanned by the new basis vectors $\begin{bmatrix} -1 & 0 \end{bmatrix}^T$ and $\begin{bmatrix} 1 & 1 \end{bmatrix}^T$, as shown in the accompanying figure.



In this case, the ratio of these areas happens to be 1.

More generally, the ratio of the volumes of the n -dimensional parallelepipeds spanned by the basis vectors of the new and old coordinate systems is called the **scaling factor** of the coordinate transformation, and it is given by the expression

$$\left| \frac{\det(A)}{\det(B)} \right|,$$

where B is the matrix containing the basis vectors of the original coordinate system and A contains the basis vectors of the new coordinate system. If the original coordinate system is the standard coordinate system, then particularly we have

$$\left| \frac{\det(A)}{\det(I)} \right| = |\det(A)|.$$

From this, we obtain the following interpretation for the determinant:

The absolute value of the determinant of matrix A , $|\det(A)|$, indicates the scaling factor of the coordinate transformation corresponding to the matrix compared to the standard coordinate system.

In fact, the sign of the determinant also has a concrete significance in this interpretation. A positive value indicates that the transformation preserves the orientation of the coordinate system, while a negative value signifies a change in orientation.

Example

Let's consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Example (continued)

Since

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \left(x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$
$$= x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$$

the matrix A keeps the vector $\mathbf{i}$ unchanged, but it maps the vectors $\mathbf{j}$ and $\mathbf{k}$ into each other. When applying the right-hand rule to the basis vectors $\mathbf{i}$ and $\mathbf{j}$ in the original coordinate system, the direction of the third basis vector $\mathbf{k}$ is obtained. If this is tested in the new coordinate system, applying the right-hand rule to the images of the vectors $\mathbf{i}$ and $\mathbf{j}$, which are now $\mathbf{i}$ and $\mathbf{k}$, we do not get the direction of the image vector $\mathbf{j}$ for vector $\mathbf{k}$, but rather the opposite direction $-\mathbf{j}$. This indicates that the orientation of the coordinate system was not preserved in the mapping. This can also be inferred from the previous result, where $\det(A) = -1 < 0$.